Graph-theory exercises

11-E01 Odd-degree vertex count

Prove that every simple graph has an even number of odd-degree vertices.

11-E02 Realize a degree sequence

Decide whether (4,3,2,2,1) is a simple graph's degree sequence. If so draw one; otherwise prove impossibility with a graphical criterion or the handshake lemma.

11-E03 Degrees and Euler traversal

A connected eight-vertex graph has exactly two degree-three vertices and all others degree four. Find its edge count and decide existence of Euler trails and circuits.

11-E04 Prove nonplanarity

Prove $K_{3,3}$ nonplanar without relying on a drawing's visible crossings.

11-E05 Colorings of C4

Derive the chromatic polynomial of C_4 using deletion–contraction or a classification of colorings.

11-E06 Bridges and Hamilton conditions

Prove a bridge cannot belong to a Hamilton cycle. Join two K_5 graphs by one bridge. Which vertices fail Dirac's condition? Explain why the ten-vertex graph has a Hamilton path but no Hamilton cycle.

11-E07 Embeddings and Euler formula

For the text's two K_4 drawings, list the common vertex and edge sets and explain why the crossing does not add a vertex. In the plane drawing delete AD, BD, AC in order; give v, e, f and prove each deleted edge is currently a nonbridge. For disjoint K_3 and K_2, compute v, e, f, c and explain why outer faces cannot be counted separately.

11-E08 Directed Euler connectivity

A digraph has arcs AB, BC, CA, AC, CD, DA, where letter order gives direction. List indegrees/outdegrees, determine Euler circuit/open trail existence, and give a full legal edge order. Add a disconnected directed triangle XYZ. What happens to degrees and why does the prior conclusion no longer follow?

11-E09 Nonplanarity of a periodic grid

A graph has vertices $(i, j) \in \mathbb{Z}_6 \times \mathbb{Z}_7$, adjacent when exactly one coordinate changes by ± 1 modulo its own modulus. Find vertex count, degree, edge count; rigorously prove triangle-freeness and use a planar bound to prove nonplanarity. Drawing crossings do not suffice.

11-E10 Three counts on labeled vertices

On five fixed labels, count simple graphs with exactly four edges, without isolates, and connected. Show intermediate formula terms and explain the difference between the last two counts.

11-E11 Vertex deletion and Hamilton paths

Prove a Hamilton path implies $c(G-S) \leq |S|+1$ for every vertex set S. Use this for $K_{2,4}$, then decide Hamilton path and cycle existence in $K_{3,4}$.

11-E12 Colorings of C5

Derive chromatic polynomials of C_3, C_4, C_5 by deletion–contraction and compute $P_{C_5}(2), P_{C_5}(3)$. Explain their relation to odd cycles and proper colorings. Do not quote a general cycle formula.

11-E13 Trace stack-based Euler traversal

For edges e1=AB, e2=BC, e3=CA, e4=CD, e5=DE, e6=EC, run the stack algorithm from A, scanning each adjacency list in increasing edge-ID order. Record S and answer after every push and pop. Why does the first pop not end the search? Give the final vertex and edge-ID sequences. Next replace EC by EF, where F is a new vertex: determine the only permissible start vertices for an open Euler trail, and explain what fails if the same code starts at A without checking its preconditions.

11-E14 Bridges and plane duals

Fix triangle ABC with leaf D drawn inside and CD as its only incident edge. Edge IDs 1..4 denote AB, BC, CA, CD. Give both boundary walks and lengths, all dual edge endpoints, a dual drawing, its vertex degrees and face count. Independently delete AB or CD from the original, keeping every vertex, and check $v - e + f = 1 + c$. If D and CD are instead drawn outside the triangle, where does the dual loop attach? Which total counts remain unchanged?

11-E15 Greedy order and the true chromatic number

The simple graph has edges $12, 23, 34$.

1. Run greedy in orders $1, 4, 2, 3$ and $1, 2, 3, 4$. At each step list prior-neighbor colors and the chosen smallest available color.
2. Give both a lower and an upper bound to determine $\chi(G)$.
3. Why does a three-color first run not prove that the graph is nonbipartite?

11-E16 Matrix powers and isomorphism invariants

1. Write A and A^2 for path $1 - 2 - 3$ and list every walk counted by $(A^2)_{22}$.
2. Compare C_6 and two disjoint triangles: vertex count, edge count, degree sequence, triangle count, and component count. Can they be isomorphic? Identify an invariant that settles the question.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study